



Qualification Pack

ASSISTANT OPERATOR- CNC TURNING - TOOL ROOM

QP Code: MSME/CSC/Q0501

Version: 1.0

NSQF Level: 3.0

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Qualification Pack

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Qualification Pack

MSME/CSC/Q0501: ASSISTANT OPERATOR- CNC TURNING -TOOL ROOM

Brief Job Description

The qualification containing different modules which is required for the job role CNC Operator- Turning, this qualifications ultimately helps learner in the following To be expertise in CNC Lathe operating cum setting

Personal Attributes

The qualification containing different modules which is required for the job role CNC Operator- Turning, this qualifications ultimately helps learner in the following To be expertise in CNC Lathe operating cum setting

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [MSME/CSC/N0502: Demonstrate the working Principle of Machine Tools](#)
2. [MSME/CSC/N0501: Assist in Operation of CNC Lathe Machine](#)
3. [MSME/CSC/N0504: Create Part Drawing using Auto-CAD](#)
4. [MSME/CSC/N0503: Checking dimensions of CNC machined Part](#)
5. [MSME/CSC/N0505: Employability Skills](#)

Qualification Pack (QP) Parameters

Sector	Capital Goods
Sub-Sector	Machine Tools
Occupation	MACHINE OPERATION 01
Country	India
NSQF Level	3.0
Credits	20
Aligned to NCO/ISCO/ISIC Code	CNC Operator - Turning



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Minimum Educational Qualification & Experience	10th grade pass with NA of experience OR Previous relevant Qualification of NSQF Level 2.5 (f NSQF Level 2.5 in metal Working/ Machine Tool area) with 1.5 years of experience OR Previous relevant Qualification of NSQF Level 2 (NSQF Level 2 in metal Working/ Machine Tool area) with 3 Years of experience
Minimum Level of Education for Training in School	10th Class
Pre-Requisite License or Training	NA
Minimum Job Entry Age	15 Years
Last Reviewed On	NA
Next Review Date	30/04/2027
NSQC Approval Date	30/04/2024
Version	1.0
Reference code on NQR	NCVET-QG-03-CG-02408-2024-V1-MSME
NQR Version	1.0



Qualification Pack

MSME/CSC/N0502: Demonstrate the working Principle of Machine Tools

Description

After completion of course Student should be able to: Understand about the basic norms of an organization

Scope

The scope covers the following :

- After completion of course Student should be able to:
- Understand about the basic norms of an organization

Elements and Performance Criteria

MSME/CCCT/02 Demonstrate the working Principle of Machine Tools

To be competent, the user/individual on the job must be able to:

- PC1.** Explain occupational health and Safety.
- PC2.** Explain about safety rules.
- PC3.**
 - State the name and location of people responsible for health and safety
 - in the workplace
- PC4.**
 - State the names and location of documents that refer to health and
 - safety in the workplace
- PC5.**
 - Using various appropriate fire extinguishers on different types of fires
 - correctly
- PC6.** Explain the PPE in Industrial Safety
- PC7.**
 - Explain Basic injury prevention, Hazard identification and avoidance,
 - safety signs for Danger, Warning, caution & personal safety message
- PC8.** Explain the types of Waste disposal techniques/ Management.
- PC9.** Explain the importance of occupational health and safety at workplace.
- PC10.** Explain the concept of 5S
- PC11.** the 5S cycle, activities and 5S program overview
- PC12.** Describe 5S program steps
- PC13.** Explain different types of machine tools (Turning machines)
- PC14.** Explain the Parts of a lathe machine
- PC15.** Explain various job holding device on lathe machine
- PC16.** Explain the function of lathe machine
- PC17.** Explain various operations performed on lathe Machine
- PC18.** Explain Methods of performing taper turning operation
- PC19.** Explain Taper turning, step turning, radius making and parting-off.
- PC20.**
 - Explain Different types of fitting tools and marking tools used in fitting
 - practice.
- PC21.** Explain the systems of unit - FPS, CGS, MKS/SI unit.
- PC22.** unit of length, Mass and time and Conversion of units



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- PC23.** • Mensuration: Area and perimeter of square, rectangle, parallelogram, triangle, circle, semi-circle, Volume of solids and cylinder .
- PC24.** • Explain quality policy and quality organization: Indian and international organization.
- PC25.** Explain the concept of quality Assurance and 7 QC Tools.



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>MSME/CCCT/02 Demonstrate the working Principle of Machine Tools</i>	100	-	-	-
PC1. Explain occupational health and Safety.	-	-	-	-
PC2. Explain about safety rules.	-	-	-	-
PC3. • State the name and location of people responsible for health and safety • in the workplace	-	-	-	-
PC4. • State the names and location of documents that refer to health and • safety in the workplace	-	-	-	-
PC5. • Using various appropriate fire extinguishers on different types of fires • correctly	-	-	-	-
PC6. Explain the PPE in Industrial Safety	-	-	-	-
PC7. • Explain Basic injury prevention, Hazard identification and avoidance, • safety signs for Danger, Warning, caution & personal safety message	-	-	-	-
PC8. Explain the types of Waste disposal techniques/ Management.	-	-	-	-
PC9. Explain the importance of occupational health and safety at workplace.	-	-	-	-
PC10. Explain the concept of 5S	-	-	-	-
PC11. the 5S cycle, activities and 5S program overview	-	-	-	-
PC12. Describe 5S program steps	-	-	-	-
PC13. Explain different types of machine tools (Turning machines)	-	-	-	-
PC14. Explain the Parts of a lathe machine	-	-	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC15. Explain various job holding device on lathe machine	-	-	-	-
PC16. Explain the function of lathe machine	-	-	-	-
PC17. Explain various operations performed on lathe Machine	-	-	-	-
PC18. Explain Methods of performing taper turning operation	-	-	-	-
PC19. Explain Taper turning, step turning, radius making and parting-off.	-	-	-	-
PC20. - Explain Different types of fitting tools and marking tools used in fitting - practice.	-	-	-	-
PC21. Explain the systems of unit - FPS, CGS, MKS/SI unit.	-	-	-	-
PC22. unit of length, Mass and time and Conversion of units	-	-	-	-
PC23. - Mensuration: Area and perimeter of square, rectangle, parallelogram, - triangle, circle, semi-circle, Volume of solids and cylinder .	-	-	-	-
PC24. - Explain quality policy and quality organization: Indian and international - organization.	-	-	-	-
PC25. Explain the concept of quality Assurance and 7 QC Tools.	-	-	-	-
NOS Total	100	-	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	MSME/CSC/N0502
NOS Name	Demonstrate the working Principle of Machine Tools
Sector	Capital Goods
Sub-Sector	
Occupation	MACHINE OPERATION 01
NSQF Level	3.0
Credits	1
Version	1.0
Last Reviewed Date	30/04/2024
Next Review Date	30/04/2027
NSQC Clearance Date	30/04/2024



Qualification Pack

MSME/CSC/N0501: Assist in Operation of CNC Lathe Machine

Description

After completion of course Student should be able to: Explain applications and advantages of CNC machines

Scope

The scope covers the following :

- After completion of course Student should be able to: Explain applications and advantages of CNC machines

Elements and Performance Criteria

MSME/CCCT/01 Assist in Operation of CNC Lathe Machine

To be competent, the user/individual on the job must be able to:

- PC1.** Explain the types of CNC Machines, Advantages & Limitations of CNC applications.
- PC2.**
 - Explain CNC interpolation, open loop & close loop control systems with feedback devices
- PC3.** Explain co-ordinate systems & points mode.
- PC4.**
 - Identify Cutting Tools and Tool Holders from the standard (ISO Standard)
- PC5.** Discuss operations of single point cutting tool
- PC6.** Selection of standard tools/ cutters/Tool Holders as per requirement
- PC7.** Explain Tool Holder Types / Styles (LH, RH and Neutral)
- PC8.** Define Turning Insert Shapes
- PC9.** Describe Operating Conditions
- PC10.** Explain Work holding methods
- PC11.** Identify and Explain Tool holding Devices
- PC12.** Explain Cutting parameters
- PC13.** Discuss the need of different oils & lubricants used
- PC14.**
 - Describe standard mathematical formulae used in calculation required for machine tool operation.
- PC15.**
 - Calculations of machining parameters like cutting speed, cutting feed, depth of cut etc.
- PC16.** Explain Coordinate System and Machine Geometry
- PC17.** Explain Axis – Orientation
- PC18.** Define Work sketch and Calculation
- PC19.**
 - Use appropriate sources to obtain the required information e.g. Numerical control on CNC machine, types of CNC control
- PC20.**
 - Check that all the equipment is correctly connected and in a safe and usable working condition



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- PC21.** • Select Appropriate Raw Material as per size of the Parts to be
• manufactured mentioned in drawing and specification
- PC22.** Calculate parameters & set a references for the various operation
- PC23.** Use appropriate techniques to create CNC program that are sufficiently and clearly detailed
- PC24.** Use codes and other references that follow the required conventions
- PC25.** Plan the machining activities before starting them.
- PC26.** Set up the suitable template/folder
- PC27.** • Set up and check that all peripheral devices are connected and
• correctly operating
- PC28.** • Confirm that the program is as per job specifications and contains all
• relevant information
- PC29.** • Make sure that programs are checked and approved by the appropriate
• person
- PC30.** Save the program in the appropriate file type and location
- PC31.** • Prepare programs, demonstrate, simulate and operate CNC lathe,
• machines for various machining operations.
- PC32.** Execute program and inspect simple geometrical forms / standard parts
- PC33.** Explain the type of maintenance and Maintenance checklist
- PC34.** Carry out Routine Maintenance activity as per standard / checklist
- PC35.** • Safe handling of tools, equipment & CNC Machines & Personal safety
• tool as per company product requirement.
- PC36.** Use protective clothing / equipment for specific tasks on CNC Machine
- PC37.** • OJT Report:
• Mentioning the process and procedure carried by the trainee for
• completing the assign task duly endorsed by the authorized personnel
• and The report must contain:
• ● Details of Department/ Organization
• ● Brief Job description & work activity
• ● Specific problem face if any with the solution.
• ● Technical Books referred during the OJT
• ● Conclusion



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>MSME/CCCT/01 Assist in Operation of CNC Lathe Machine</i>	-	100	-	-
PC1. Explain the types of CNC Machines, Advantages & Limitations of CNC applications.	-	-	-	-
PC2. • Explain CNC interpolation, open loop & close loop control systems with • feedback devices	-	-	-	-
PC3. Explain co-ordinate systems & points mode.	-	-	-	-
PC4. • Identify Cutting Tools and Tool Holders from the standard (ISO • Standard)	-	-	-	-
PC5. Discuss operations of single point cutting tool	-	-	-	-
PC6. Selection of standard tools/ cutters/Tool Holders as per requirement	-	-	-	-
PC7. Explain Tool Holder Types / Styles (LH, RH and Neutral)	-	-	-	-
PC8. Define Turning Insert Shapes	-	-	-	-
PC9. Describe Operating Conditions	-	-	-	-
PC10. Explain Work holding methods	-	-	-	-
PC11. Identify and Explain Tool holding Devices	-	-	-	-
PC12. Explain Cutting parameters	-	-	-	-
PC13. Discuss the need of different oils & lubricants used	-	-	-	-
PC14. • Describe standard mathematical formulae used in calculation required • for machine tool operation.	-	-	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC15. • Calculations of machining parameters like cutting speed, cutting feed, • depth of cut etc.	-	-	-	-
PC16. Explain Coordinate System and Machine Geometry	-	-	-	-
PC17. Explain Axis - Orientation	-	-	-	-
PC18. Define Work sketch and Calculation	-	-	-	-
PC19. • Use appropriate sources to obtain the required information e.g. • Numerical control on CNC machine, types of CNC control	-	-	-	-
PC20. • Check that all the equipment is correctly connected and in a safe and • usable working condition	-	-	-	-
PC21. • Select Appropriate Raw Material as per size of the Parts to be • manufactured mentioned in drawing and specification	-	-	-	-
PC22. Calculate parameters & set a references for the various operation	-	-	-	-
PC23. Use appropriate techniques to create CNC program that are sufficiently and clearly detailed	-	-	-	-
PC24. Use codes and other references that follow the required conventions	-	-	-	-
PC25. Plan the machining activities before starting them.	-	-	-	-
PC26. Set up the suitable template/folder	-	-	-	-
PC27. • Set up and check that all peripheral devices are connected and • correctly operating	-	-	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC28. Confirm that the program is as per job specifications and contains all relevant information	-	-	-	-
PC29. Make sure that programs are checked and approved by the appropriate person	-	-	-	-
PC30. Save the program in the appropriate file type and location	-	-	-	-
PC31. - Prepare programs, demonstrate, simulate and operate CNC lathe, - machines for various machining operations.	-	-	-	-
PC32. Execute program and inspect simple geometrical forms / standard parts	-	-	-	-
PC33. Explain the type of maintenance and Maintenance checklist	-	-	-	-
PC34. Carry out Routine Maintenance activity as per standard / checklist	-	-	-	-
PC35. - Safe handling of tools, equipment & CNC Machines & Personal safety - tool as per company product requirement.	-	-	-	-
PC36. Use protective clothing / equipment for specific tasks on CNC Machine	-	-	-	-
PC37. - OJT Report: - Mentioning the process and procedure carried by the trainee for - completing the assign task duly endorsed by the authorized personnel - and The report must contain: ● Details of Department/ Organization ● Brief Job description & work activity ● Specific problem face if any with the solution. ● Technical Books referred during the OJT ● Conclusion	-	-	-	-
NOS Total	-	100	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	MSME/CSC/N0501
NOS Name	Assist in Operation of CNC Lathe Machine
Sector	Capital Goods
Sub-Sector	
Occupation	MACHINE OPERATION 01
NSQF Level	3.0
Credits	14
Version	1.0
Last Reviewed Date	30/04/2024
Next Review Date	30/04/2027
NSQC Clearance Date	30/04/2024



Qualification Pack

MSME/CSC/N0504: Create Part Drawing using Auto-CAD

Description

After completion of course Student should be able to Explain the application of engineering drawing.

Scope

The scope covers the following :

- After completion of course Student should be able to
- Explain the application of engineering drawing.

Elements and Performance Criteria

MSME/CCCT/04 Create Part Drawing using Auto-CAD

To be competent, the user/individual on the job must be able to:

- PC1.** Explain the Importance of Engineering drawing,
- PC2.** Explanation the scope and objective of Engineering Drawing
- PC3.**
 - Demonstrate and explain drawing Standards: Size of drawing sheets -
 - Layout of drawing sheet - Title Blocks - Types of lines - Folding of
 - drawing sheets.
- PC4.** Use of dimensioning techniques according to Standard of dimensions
- PC5.**
 - Demonstrate orthographic & Isometric projection by using a viewing
 - box and a model
- PC6.** Demonstrate first angle and third angle projection
- PC7.** Use of symbol in projections -Front view, top view and side view
- PC8.** Demonstrate the use of Auto CAD and Auto CAD interface
- PC9.** Apply coordinates systems in auto CAD
- PC10.** Demonstrate the use of tool bars.
- PC11.** Create solid field area (Hatching, Gradient)
- PC12.** Edit objects using the object property tool bar and various methods.
- PC13.** Use sketch settings and Style toolbar (text style, Multilayer style etc.)
- PC14.** Edit object using object property toolbar & various method.
- PC15.** Create the replica of model using copy, array command
- PC16.** Work with models in the modify toolbar.
- PC17.** Identify the appropriate Tool to create and modify the model
- PC18.** Change the orientation of the object by aligns, offset, rotate command
- PC19.** Apply standard dimension in a mechanical component.
- PC20.**
 - Use of dimensioning Methods: Linear, Align, ordinates, Radius,
 - Diameter, Arc length, angular etc
- PC21.** Use of leader with text, block reference
- PC22.** Edit or modify the CAD Drawings



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- PC23.** Use of layers Management and its applications
- PC24.** Apply GD& T Symbols in drawings
- PC25.** Develop proper drawing layout.



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>MSME/CCCT/04 Create Part Drawing using Auto-CAD</i>	-	100	-	-
PC1. Explain the Importance of Engineering drawing,	-	-	-	-
PC2. Explanation the scope and objective of Engineering Drawing	-	-	-	-
PC3. • Demonstrate and explain drawing Standards: Size of drawing sheets - • Layout of drawing sheet - Title Blocks - Types of lines - Folding of • drawing sheets.	-	-	-	-
PC4. Use of dimensioning techniques according to Standard of dimensions	-	-	-	-
PC5. • Demonstrate orthographic & Isometric projection by using a viewing • box and a model	-	-	-	-
PC6. Demonstrate first angle and third angle projection	-	-	-	-
PC7. Use of symbol in projections -Front view, top view and side view	-	-	-	-
PC8. Demonstrate the use of Auto CAD and Auto CAD interface	-	-	-	-
PC9. Apply coordinates systems in auto CAD	-	-	-	-
PC10. Demonstrate the use of tool bars.	-	-	-	-
PC11. Create solid field area (Hatching, Gradient)	-	-	-	-
PC12. Edit objects using the object property tool bar and various methods.	-	-	-	-
PC13. Use sketch settings and Style toolbar (text style, Multilayer style etc.)	-	-	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. Edit object using object property toolbar & various method.	-	-	-	-
PC15. Create the replica of model using copy, array command	-	-	-	-
PC16. Work with models in the modify toolbar.	-	-	-	-
PC17. Identify the appropriate Tool to create and modify the model	-	-	-	-
PC18. Change the orientation of the object by aligns, offset, rotate command	-	-	-	-
PC19. Apply standard dimension in a mechanical component.	-	-	-	-
PC20. • Use of dimensioning Methods: Linear, Align, ordinates, Radius, • Diameter, Arc length, angular etc	-	-	-	-
PC21. Use of leader with text, block reference	-	-	-	-
PC22. Edit or modify the CAD Drawings	-	-	-	-
PC23. Use of layers Management and its applications	-	-	-	-
PC24. Apply GD& T Symbols in drawings	-	-	-	-
PC25. Develop proper drawing layout.	-	-	-	-
NOS Total	-	100	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	MSME/CSC/N0504
NOS Name	Create Part Drawing using Auto-CAD
Sector	Capital Goods
Sub-Sector	
Occupation	MACHINE OPERATION 01
NSQF Level	3.0
Credits	3
Version	1.0
Last Reviewed Date	30/04/2024
Next Review Date	30/04/2027
NSQC Clearance Date	30/04/2024



Qualification Pack

MSME/CSC/N0503: Checking dimensions of CNC machined Part

Description

After completion of course Student should be able to Understand End and line Standard

Scope

The scope covers the following :

- After completion of course Student should be able to
- Understand End and line Standard

Elements and Performance Criteria

MSME/CCCT/03 Checking dimensions of CNC machined Part

To be competent, the user/individual on the job must be able to:

- PC1.** • Demonstrate the linear measurement by vernier Caliper (Digital, Analog and Dial type)
- PC2.** • Demonstrate the linear measurement by micrometer (Analog and Digital type)
- PC3.** • Demonstrate the linear Measurement by height gauges (Digital and Analog type).
- PC4.** • Calculate the least count of various instruments (Vernier Caliper, Micrometer and Height Gauge)
- PC5.** • Demonstrate the diametrical Measurement: inside diameter by inside micrometer
- PC6.** • Demonstrate the Diametrical Measurement: depth of hole or recess by depth micrometer.
- PC7.** Angle and taper measurements by bevel protractor and Sine bar.
- PC8.** • To find out the flatness of Surface plate by use of spirit level and optical flat.
- PC9.** Measure surface roughness by roughness tester.
- PC10.** • Perform measurement of micro threads (British , Acme, Metric) by use of profile projector
- PC11.** Apply limits, fits and tolerances for a given geometry
- PC12.** Check various dimensions using gauges (Go-No GO Type, plug, ring etc.)
- PC13.** Measurement of screw thread using various instruments
- PC14.** • List Geometrical Dimensions & Tolerances used in manufacturing industries.
- PC15.** Evaluate and do analysis of parameters of screw threads Measurement:



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>MSME/CCCT/03 Checking dimensions of CNC machined Part</i>	-	100	-	-
PC1. • Demonstrate the linear measurement by vernier Caliper (Digital, • Analog and Dial type)	-	-	-	-
PC2. • Demonstrate the linear measurement by micrometer (Analog and • Digital type)	-	-	-	-
PC3. • Demonstrate the linear Measurement by height gauges (Digital and • Analog type).	-	-	-	-
PC4. • Calculate the least count of various instruments (Vernier Caliper, • Micrometer and Height Gauge)	-	-	-	-
PC5. • Demonstrate the diametrical Measurement: inside diameter by inside • micrometer	-	-	-	-
PC6. • Demonstrate the Diametrical Measurement: depth of hole or recess by • depth micrometer.	-	-	-	-
PC7. Angle and taper measurements by bevel protractor and Sine bar.	-	-	-	-
PC8. • To find out the flatness of Surface plate by use of spirit level and optical • flat.	-	-	-	-
PC9. Measure surface roughness by roughness tester.	-	-	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. - Perform measurement of micro threads (British , Acme, Matric) by use - of profile projector	-	-	-	-
PC11. Apply limits, fits and tolerances for a given geometry	-	-	-	-
PC12. Check various dimensions using gauges (Go-No GO Type, plug, ring etc.)	-	-	-	-
PC13. Measurement of screw thread using various instruments	-	-	-	-
PC14. - List Geometrical Dimensions & Tolerances used in manufacturing - industries.	-	-	-	-
PC15. Evaluate and do analysis of parameters of screw threads Measurement:	-	-	-	-
NOS Total	-	100	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	MSME/CSC/N0503
NOS Name	Checking dimensions of CNC machined Part
Sector	Capital Goods
Sub-Sector	
Occupation	MACHINE OPERATION 01
NSQF Level	3.0
Credits	1
Version	1.0
Last Reviewed Date	30/04/2024
Next Review Date	30/04/2027
NSQC Clearance Date	30/04/2024



Qualification Pack

MSME/CSC/N0505: Employability Skills

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and getting ready for jobs and apprenticeship.

Scope

The scope covers the following :

- This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century,
- digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service,
- entrepreneurship, and getting ready for jobs and apprenticeship.

Elements and Performance Criteria

MSME/ES/01 Employability skills

To be competent, the user/individual on the job must be able to:

- PC1.** Explain the major applications of MS Office
- PC2.** Explain the different types of e-commerce
- PC3.** List the benefits of e-commerce for retailers and customers
- PC4.**
 - Discuss how the Digital India campaign will help boost e-commerce in India
- PC5.** Write applications pertaining to various matters.
- PC6.** Explain power of positive attitude and Importance of commitment
- PC7.**
 - Explain motivation and the Ways to motivate oneself and Personal goal
 - setting
- PC8.** Explain the Effective & Level of Communication
- PC9.** Explain communication and Significance of technical communication
- PC10.** Explain the methods of listening Skills.
- PC11.** Explain the differences between bio-data, CV and Resume.
- PC12.** Explain verbal and non-verbal Communication
- PC13.** Explain how to face an interview
- PC14.** Explain team work, group work, team formation process
- PC15.** How to Minimize the team conflicts
- PC16.** Explain Ethics & values
- PC17.**
 - Explain the concept of entrepreneurship, and entrepreneurship v/s Management
- PC18.**
 - Explain the process of project report preparation for setting up a new business



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- PC19.** • Explain the role of various schemes and institute for self-employment
• i.e MSME, DIC, NSIC, SIDBI etc
- PC20.** Role of financial institution to support startup
- PC21.** Discuss the importance of saving money
- PC22.** Discuss the main types of bank accounts
- PC23.** Differentiate between fixed and variable costs
- PC24.** Describe the different types of insurance products
- PC25.** Discuss the main types of electronic funds transfers



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>MSME/ES/01 Employability skills</i>	97	-	-	-
PC1. Explain the major applications of MS Office	-	-	-	-
PC2. Explain the different types of e-commerce	-	-	-	-
PC3. List the benefits of e-commerce for retailers and customers	-	-	-	-
PC4. • Discuss how the Digital India campaign will help boost e-commerce in • India	-	-	-	-
PC5. Write applications pertaining to various matters.	-	-	-	-
PC6. Explain power of positive attitude and Importance of commitment	-	-	-	-
PC7. • Explain motivation and the Ways to motivate oneself and Personal goal • setting	-	-	-	-
PC8. Explain the Effective & Level of Communication	-	-	-	-
PC9. Explain communication and Significance of technical communication	-	-	-	-
PC10. Explain the methods of listening Skills.	-	-	-	-
PC11. Explain the differences between bio-data, CV and Resume.	-	-	-	-
PC12. Explain verbal and non-verbal Communication	-	-	-	-
PC13. Explain how to face an interview	-	-	-	-
PC14. Explain team work, group work, team formation process	-	-	-	-
PC15. How to Minimize the team conflicts	-	-	-	-



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC16. Explain Ethics & values	-	-	-	-
PC17. - Explain the concept of entrepreneurship, and entrepreneurship v/s - Management	-	-	-	-
PC18. - Explain the process of project report preparation for setting up a new - business	-	-	-	-
PC19. - Explain the role of various schemes and institute for self- employment - i.e MSME, DIC, NSIC, SIDBI etc	-	-	-	-
PC20. Role of financial institution to support startup	-	-	-	-
PC21. Discuss the importance of saving money	-	-	-	-
PC22. Discuss the main types of bank accounts	-	-	-	-
PC23. Differentiate between fixed and variable costs	-	-	-	-
PC24. Describe the different types of insurance products	-	-	-	-
PC25. Discuss the main types of electronic funds transfers	-	-	-	-
NOS Total	100	-	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	MSME/CSC/N0505
NOS Name	Employability Skills
Sector	Capital Goods
Sub-Sector	
Occupation	MACHINE OPERATION 01
NSQF Level	3
Credits	1
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

as per QP

Minimum Aggregate Passing % at QP Level : 40

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Minimum Passing % at NOS Level: 40

(Please note: A Trainee must score the minimum percentage for each NOS separately as well as on the QP as a whole.)

Assessment Weightage

Compulsory NOS



Qualification Pack

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
MSME/CSC/N0502.Demonstrate the working Principle of Machine Tools	100	-	-	-	100	10
MSME/CSC/N0501.Assist in Operation of CNC Lathe Machine	-	100	-	-	100	60
MSME/CSC/N0504.Create Part Drawing using Auto-CAD	-	100	-	-	100	10
MSME/CSC/N0503.Checking dimensions of CNC machined Part	-	100	-	-	100	10
MSME/CSC/N0505.Employability Skills	100	-	-	-	100	10
Total	200	300	-	-	500	100



Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training



Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.



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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.